4.The online train reservation system allows users to search for trains by route, date, and time, and view available seats and classes. Users can register, log in, and manage their reservations through a personal dashboard. The booking process involves selecting seats, entering passenger details, and making secure payments via an integrated payment gateway. Admins can manage train schedules, seat availability, and user accounts through a comprehensive admin panel. Users receive booking confirmation emails and can view their booking history and print tickets from their dashboard. The website features real-time seat availability updates and is fully responsive for seamless use on all devices. The Reservation system should contain the following feature

1. Aim

To design and develop an Online Train Reservation System using HTML, CSS and JavaScript that allows users to search trains, select seats, enter passenger details and confirm bookings**.**

2. Algorithm

1. Start the program.
2. Display the train reservation homepage.
3. Allow users to register and log in.
4. Enter source, destination and journey date.
5. Display available trains, timings and ticket classes.
6. Allow users to select a train and available seats.
7. Enter passenger details.
8. Calculate the ticket fare and proceed with payment.
9. Display booking confirmation and reservation details.
10. Allow users to view booking history and print tickets.
11. Admin can manage train schedules and seat availability.
12. Stop the program.

HTMLCODE:

```
<!DOCTYPE html>
<html>
<head>
<title>Online Train Reservation</title>
<style>
body {
    font-family: Arial;
    background: #eef2ff;
    margin: 0;
}
header {
    background: #1e40af;
    color: white;
    padding: 20px;
    text-align: center;
}
.box {
    background: white;
    padding: 25px;
    margin: 25px auto;
    width: 80%;
    border-radius: 10px;
}
input, select {
    width: 95%;
    padding: 10px;
    margin: 8px;
}
button {
```

```
    background: #1e40af;
    color: white;
    padding: 12px;
    border: none;
    cursor: pointer;
    margin: 5px;
}
</style>
</head>

<body>

<header>
<h1>ONLINE TRAIN RESERVATION</h1>
<p>Book Your Train Tickets Easily</p>
</header>

<div class="box">
<h2>User Login</h2>
<input type="text" placeholder="Username">
<input type="password" placeholder="Password">
<button onclick="alert('Login Successful')">Login</button>
</div>

<div class="box">
<h2>Search Trains</h2>

<input type="text" id="from" placeholder="From Station">
<input type="text" id="to" placeholder="To Station">
<input type="date" id="date">
```

```
<button onclick="searchTrain()">Search Train</button>

<p id="train"></p>
</div>

<div class="box">
<h2>Seat Reservation</h2>

<select id="seat">
<option value="Sleeper">Sleeper</option>
<option value="AC 3 Tier">AC 3 Tier</option>
<option value="AC 2 Tier">AC 2 Tier</option>
</select>

<input type="text" id="name" placeholder="Passenger Name">
<input type="number" id="age" placeholder="Passenger Age">

<button onclick="bookTicket()">Book Ticket</button>

<p id="result"></p>
<button onclick="window.print()">Print Ticket</button>
</div>

<div class="box">
<h2>Admin Panel</h2>
<p>Manage train schedules and seat availability.</p>
<button onclick="alert('Admin Panel Opened')">Manage Trains</button>
</div>
```

```
<script>
function searchTrain() {
    let from = document.getElementById("from").value;
    let to = document.getElementById("to").value;

    if(from == "" || to == "") {
        alert("Enter source and destination");
    } else {
        document.getElementById("train").innerHTML =
        "Train: Express 12601<br>" +
        "Route: " + from + " to " + to +
        "<br>Available Seats: 25";
    }
}

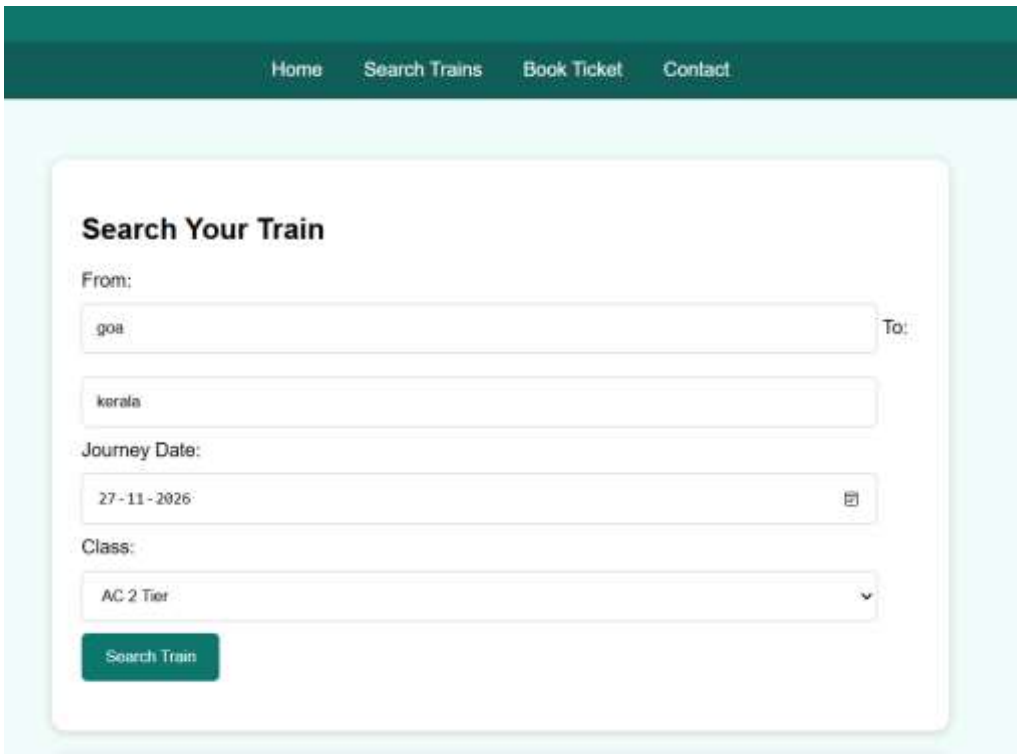
function bookTicket() {
    let name = document.getElementById("name").value;
    let age = document.getElementById("age").value;
    let seat = document.getElementById("seat").value;

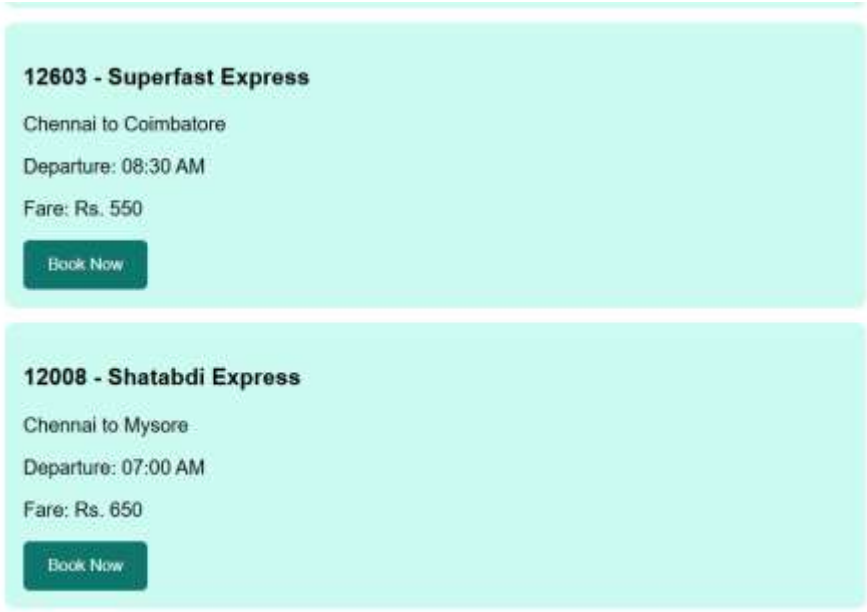
    if(name == "" || age == "") {
        alert("Enter passenger details");
    } else {
        document.getElementById("result").innerHTML =
        "Booking Confirmed!<br>" +
        "Passenger: " + name +
        "<br>Age: " + age +
        "<br>Class: " + seat +
        "<br>Payment: Sample Payment Successful";
    }
```

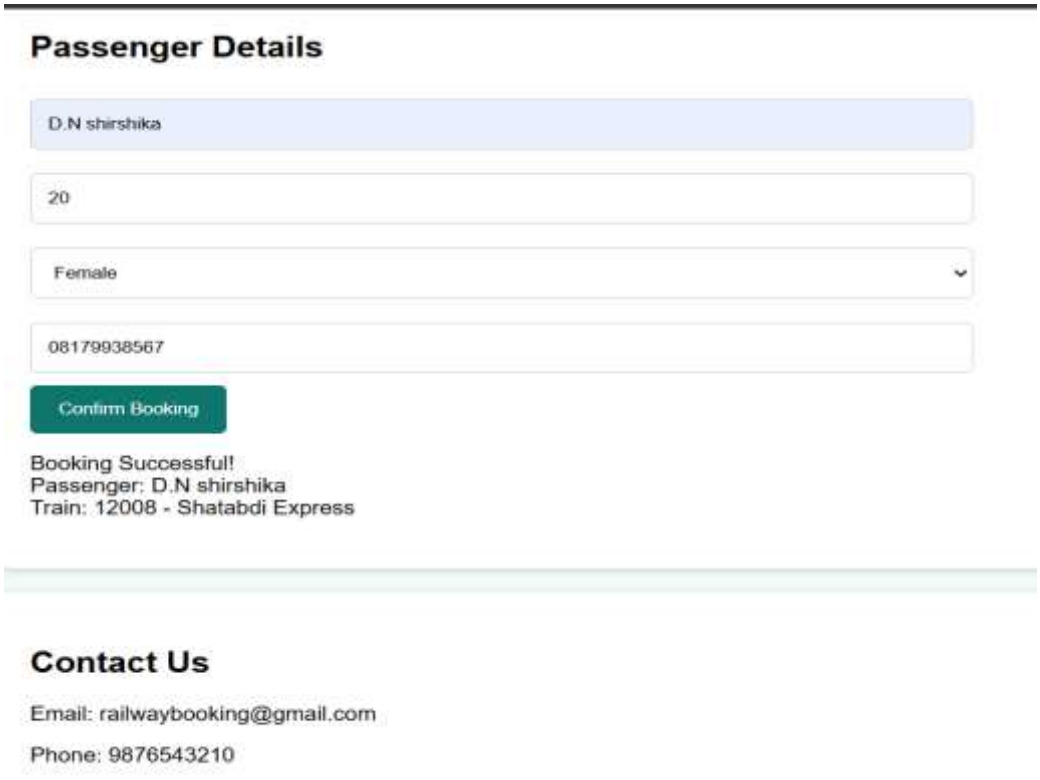
```
}
</script>

</body>
</html>
```

Output







Result

Thus, the Online Train Reservation System was designed using HTML, CSS and JavaScript. It allows users to search trains, enter passenger details, select travel classes and display booking confirmation.